

B2G TURNKEY SOLUTION

# Climate-Resilient Urban Forestry

A comprehensive, cloud-integrated hardware and software ecosystem delivering actionable soil intelligence to city administrations and public communities.



# Our Municipal Service Ecosystem



## 1. ANALYTICS

### Cloud Platform

Advanced starlight & satellite image map processing integrated into central municipal dashboards.



## 2. HARDWARE

### MCU & Sensors

Sealed ESP32 edge units measuring ground-truth hydration depths natively inside urban street soils.



## 3. FIELD APP

### Technician Portal

Mobile-optimized field client built for rapid calibration, testing, and hot-swap configuration.



## 4. COMMUNITY

### Citizen App

Voluntary contribution hub gamifying citizen-led watering and environmental data collection.



# 1. Cloud Platform & GIS Mapping

## MACRO MONITORING

### Satellite & Map Processing

By blending micro-sensor ground data with orbital satellite imaging, we construct a living model of the city's green health:

-  **Starlight Imagery:** Advanced visual processing channels to monitor canopy heat stress patterns across entire districts.
-  **GIS Map Layering:** Overlays local topology and soil permeability indexes to predict urban water pooling trends.
-  **Dynamic Hydration Mapping:** Generates real-time, high-priority municipal driving paths for water tanker trucks.

### Satellite smart city map dashboard



## 2. Resilient IoT Hardware Nodes

### MICRO SENSING

#### Soil-Grade MCU Nodes

Industrial-tier, low-profile telemetry hardware built to collect precise ground metrics under asphalt environments:

- 🔌 **ESP32 Microcontroller:** Low-cost, highly scalable system architecture running optimized, secure deep sleep firmware.
- 💧 **Capacitive Sensors:** Corrosion-free moisture probes measuring raw dielectric permittivity to determine true root hydration.
- 🛡️ **IP68 Structural Vaults:** Protected enclosures designed for long-term placement below soil surfaces, safe from public vandalism.





## 3. Staff Portal vs 4. Citizen App

### FOR CITY CREWS

#### Field Tech Portal

##### Rapid Ground Configuration

A highly responsive, stripped-down web app used by municipal field crews to deploy, test, and troubleshoot devices directly on-site.

 One-tap sensor calibration (Dry vs Wet threshold map).

 Real-time LoRaWAN signal testing and configuration.

### FOR PUBLIC VOLUNTEERS

#### Community App

##### Civic Volunteer Engagement

A public-facing platform designed to empower neighborhood groups to coordinate hydration tasks on a structured, active scale.

 Sponsor a local street tree and unlock public badges.

 Instant logging of community watering events via QR.



# Partnership Offer

Driving Cost Efficiency and Civic Trust in Municipal Infrastructure

 [partnerships@smartcanopy-b2g.com](mailto:partnerships@smartcanopy-b2g.com) |  [www.smartcanopy-b2g.com](http://www.smartcanopy-b2g.com)



# Image Sources



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